the position, orientation and movement of the body. The organ of balance is known as the **vestibular system** which is located in the inner ear next to the cochlea. The vestibular system is composed of a series of fluid filled sacs and tubules.These sacs and tubules contain endolymph and are kept in the surrounding perilymph (Figure 10.19). These two fluids, perilymph and endolymph, respond to the mechanical forces, during changes occurring in body position and acceleration.

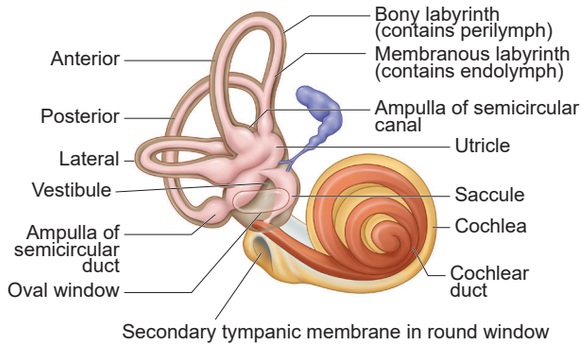


Figure 10.19 Organ of Equilibrium

Name the parts of the organ of equilibrium involved in the following functions.

a) Linear movement of the body

b) Changes in the body position

c) Rotational movement of the head.

The utricle and saccule are two membranous sacs, found nearest the cochlea and contain equilibrium receptor regions called **maculae** that are involved in detecting the linear movement of the head. The maculae contain the hair cells that act as mechanorecptors. These hair cells are embeded in a gelatinous otolithic membrane that contains small calcareous particles called **otoliths**. This membrane adds weight to the top of the hair cells and increase the inertia.

The canals that lie posterior and lateral to the vestibule are semicircular canals; they are **anterior, posterior and lateral canals** oriented at right angles to each other**.** At one end of each semicircular canal, at its lower end has a swollen area called **ampulla**. Each ampulla has a sensory area known as **crista ampullaris** which is formed of sensory hair cells and supporting cells. The function of these canals is to detect rotational movement of the head.

The intensity of sound is measured in decibels (dB). 0 dB is the threshold of hearing for normal ear. Severe hearing loss occurs with frequent or prolonged exposure to sound with intensities greater than 90dB. For meaningful conversations the intensity should be in the 50 dB range.

10.6.3 Olfactory receptors

The receptors for taste and smell are the chemoreceptors. The smell receptors are excited by air borne chemicals that dissolve in fluids. The yellow coloured patches of olfactory epithelium form the olfactory organs that are located on the roof of the nasal cavity. The olfactory epithelium is covered by a thin coat of mucus layer below and olfactory glands bounded connective tissues, above. It contains three types of cells**: supporting cells, Basal cells** and millions of pin shaped **olfactory receptor cells** (which are unusual bipolar cells). The olfactory glands and the supporting cells secrete the mucus. The unmyelinated axons of the olfactory receptor cells are gathered to form the filaments of olfactory nerve [cranial nerve I] which synapse with cells of olfactory bulb. The impulse, through the olfactory nerves, is transmitted to the frontal

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